is loosened and the force acting on the brake drum is released. Due to this action the wheels connected with the brake drums are started to rotate without any difficulties. Figure 6.2 shows the view of an externally contracting type brake used in cars.

6.4 Power brake

In mechanical braking system, when the braking force is applied by the driver by pressing the brake pedal due to the mechanical advantage combined with braking force the vehicle is stopped. However, when the vehicle's weight and speed are increased, the mechanical braking system will be no longer capable of providing sufficient braking force to stop the vehicle immediately. Hence for providing sufficient power to the mechanical braking system and to increase the braking force we use any one of the following support as a supplemental force. This type of brake is called as power brake. This power brake is classified according to the following categories as

1. Hydraulic brake
2. Air brake
3. Vacuum brake
4. Electric brake
5. Servo brake

6.4.1 Hydraulic brake system

The schematic view of the hydraulic braking system can be seen in Figure 6.4.1.

The hydraulic braking system works based on the principle of Pascal's law. It is simple in construction and can provide more braking force. The hydraulic brake is capable of delivering the braking

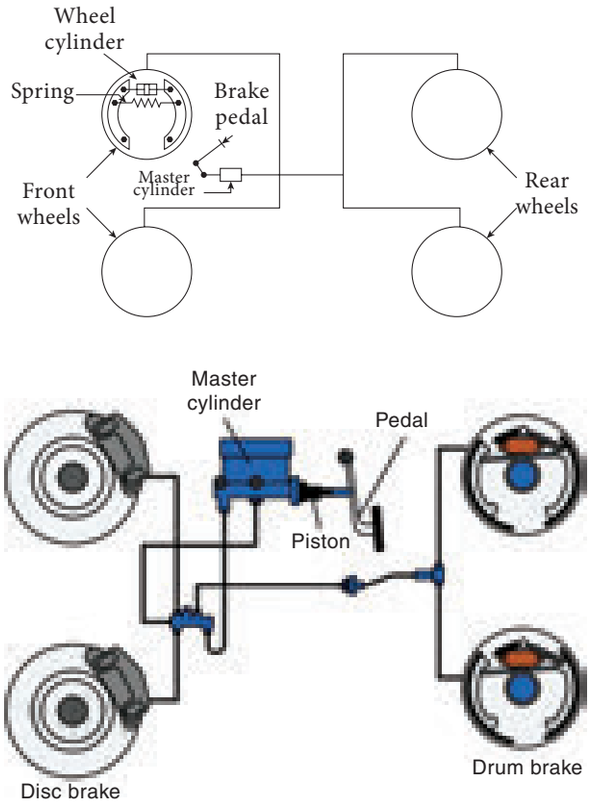


Figure 6.4.1 The schematic view of the hydraulic braking system

force evenly to all the wheels at a time, smoothly and consistently for all the wheels. Therefore most of the cars and some of the two wheelers are employed with hydraulic braking system.

Construction and working principle of hydraulic brake system

The construction of the hydraulic brake system could refered from the Figure 6.3. The hydraulic brake system consists of a master cylinder and wheel cylinders which are the two important components of the system. The master cylinder is used to store the brake fluid. One side of the master cylinder is connected with brake pedal and on the other side there is a pipe line which connects the wheel

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